



Call Insights Pipeline

AI-powered call analysis for sales teams

Turning every sales call into structured, scored CRM data — automatically.

The Problem

Why call data dies after the call ends



Calls go unreviewed

Hundreds of recordings pile up in storage. Managers can't listen to them all. Insights stay locked in audio.



Manual notes are inconsistent

Counselors log calls differently — or skip logging entirely. CRM data is incomplete and biased.



Lead quality is invisible

No one can quickly tell which calls actually moved a lead forward versus a polite chat with no outcome.

The Solution

A pipeline that listens, understands, and writes to CRM — on its own

Every call, automatically:

- Downloaded & stored permanently
- Transcribed (English, Hindi, Telugu)
- Speakers labelled (agent vs customer)
- Scored 0–100 on lead quality
- Pushed to Zoho CRM in seconds



Call recorded



Whisper transcribes



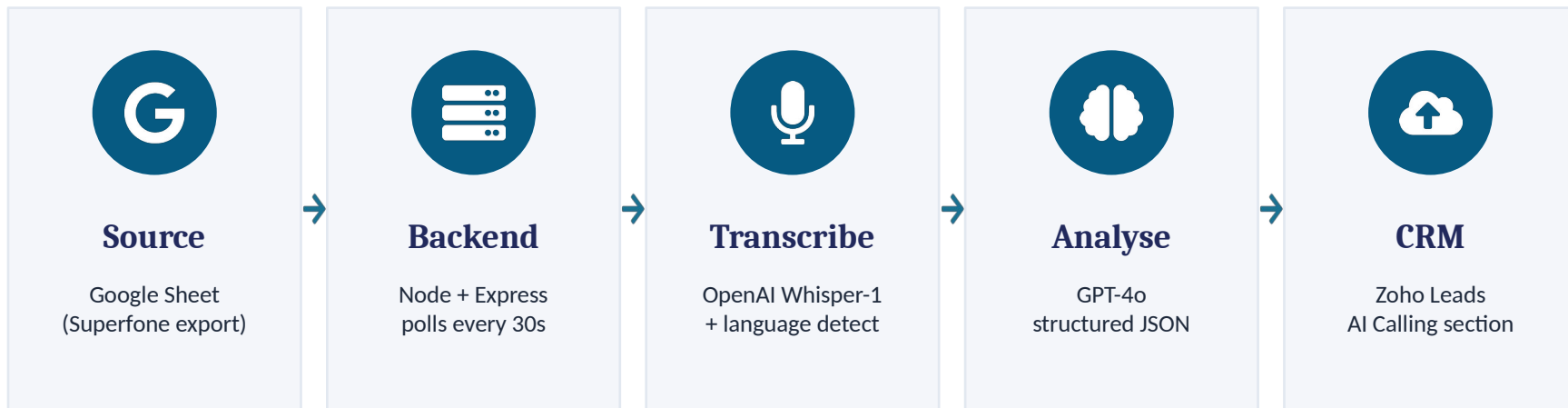
GPT-4o analyses



Zoho CRM updated

System Architecture

End-to-end data flow from recording to CRM



Storage & state:



Supabase Postgres — calls, batches, transcripts



Supabase Storage — permanent MP3 archive

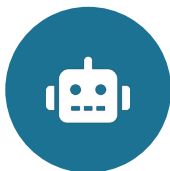
Tech Stack

Production-grade tools, self-hosted backend



Node.js + Express

Async backend, retry logic, structured logging



Whisper-1

OpenAI multilingual speech-to-text (EN / HI / TE)



GPT-4o

OpenAI structured analysis + speaker labelling



Supabase

Postgres DB + object storage for audio



Google Sheets API

Live polling of Superfone call log



Zoho CRM

OAuth integration, custom field mapping

What the AI Extracts

From a single call recording, the system produces 12+ structured fields

Score (0–100)	Calibrated lead-quality rating	Call status	Answered / Missed / Busy / Not Connected
Quality category	High Quality / Average / Poor / Very Poor	Follow-up required?	True if counselor promised callback
Sentiment	Positive / Neutral / Negative	College interested?	Boolean signal for funnel stage
Interest level	High / Medium / Low / None	AI Tags	walkin_interested, high_score, etc.
Program & course	Intermediate, B.Tech MPC, CGA, MEC, etc.	Summary	2–3 sentence call abstract
Walk-in interested?	Boolean + parsed date if confirmed	Full transcript	Speaker-labelled dialogue

Engineering Highlights

Decisions that keep the pipeline reliable in production



Phase-1 safe Zoho push

Whitelist + dry-run + upsert-by-phone. Never writes garbage leads to live CRM. Robust matching across Phone and Mobile fields, 6 number formats tried.



UUID-based deduplication

Real per-call IDs extracted from S3 URLs replaced the unreliable SF Number column. Postgres unique index acts as a hard safety net.



Retry + fail-soft pipeline

Every OpenAI call wrapped in 3-attempt retry with backoff. Speaker labelling falls back to raw transcript if it stalls — analysis never blocks.

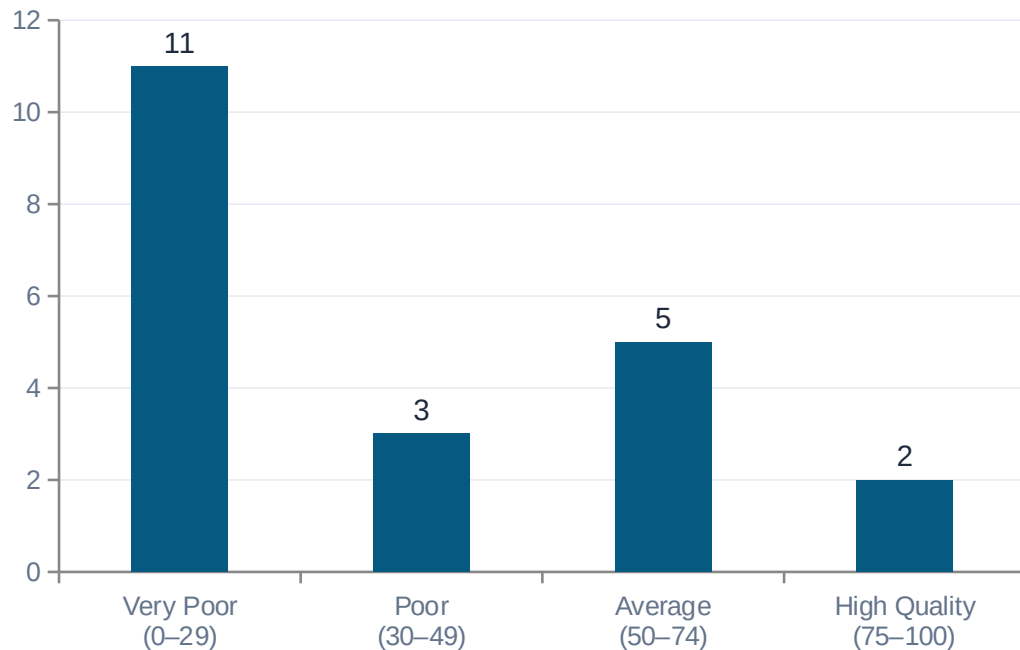


Cost-aware short-circuits

Calls under 10 seconds skip Whisper entirely. Expired S3 URLs detected from XML response body. Saves credits and reduces noise.

Sample Run — Score Distribution

Real test batch: how the scoring rubric separates call quality



Key insight

52%

of calls scored "Very Poor" — failed dials, voicemails, wrong numbers.

Without AI scoring, these would sit in CRM looking identical to real conversations.

Real Output Example

A 90-score call analysed in under 2 minutes, end-to-end

INPUT

Audio recording — 3 min 12 s

Counselor Roja → Customer Haritha

Mixed-language call. Customer's child has completed 10th class, no clarity on stream. Counselor explains psychometric assessment, available programs, and books a campus visit.

AI OUTPUT

90 / 100

High Quality • Positive sentiment

Program	Intermediate
Walk-in	Confirmed — 29 May, Jubilee Hills
Interest	High
Tags	walkin_interested, high_score

Status & Roadmap

What's live today and what's coming next

Shipped

- ✓ Whisper transcription with retry
- ✓ Speaker-labelled dialogue formatting
- ✓ GPT-4o scoring with custom rubric
- ✓ Supabase storage for permanent audio
- ✓ Google Sheet polling every 30 s
- ✓ Zoho upsert + create with safe guardrails
- ✓ UUID dedup with unique-index safety net

Next up

- ➔ Speaker diarization at audio level (replace text-based guessing)
- ➔ Auto-trigger workflows on score thresholds (e.g. notify manager on > 80)
- ➔ Dashboard analytics — agent leaderboards, weekly trends
- ➔ Multi-tenant support — same backend, many client orgs
- ➔ WhatsApp / SMS follow-up automation for walk-in confirmations



**From audio to action,
in under two minutes.**

github.com/JashvithaS/call-insights-pipeline

T h a n k y o u .